

Missouri S&T
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Section 4.3

Linearly Independent Sets

Bases

Linearly Independent Sets

An indexed set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ of vectors in a vector space V is linearly independent if the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution $c_1 = \dots = c_p = 0$.

Linearly Dependent Sets

An indexed set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ of vectors in a vector space V is linearly dependent if the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p = \mathbf{0}$$

has a nontrivial solution where at least one of the weights $c_1, \dots, c_p$ is nonzero.

In this case, the vector equation above provides a linear dependence relation among the vectors.

Comments on Linear (In)dependence

Any set containing the zero vector is linearly dependent.

A set containing only a single vector $\mathbf{v}$ is linearly independent if and only if $\mathbf{v} \neq \mathbf{0}$.

A set of two vectors is linearly dependent if and only if one of the vectors is a multiple of the other.

Theorem

An indexed set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ of two or more vectors (all of which are nonzero) is linearly dependent if and only if some $\mathbf{v}_j$ (with $j > 1$) is a linear combination of the preceding vectors.

Example 1

Consider the vector space $C[0,1]$, the space of all continuous functions on $0 \leq t \leq 1$.

Identify which of the following sets are linearly independent and which are linearly dependent.

- $\{\sin t, \cos t\}$
- $\{\sin t, \cos t, \sin 2t\}$
- $\{\sin t \cos t, \sin 2t\}$

A Basis for a Vector Space

Let H be a subspace of a vector space V . An indexed set of vectors $B = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ in V is a basis for H if

- i. B is a linearly independent set, and
- ii. the subspace spanned by B is H ; that is,
$$H = \text{Span}\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$$

The plural of basis is bases.

Note that, trivially, V is a subspace of V .

Example 2

Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ be an invertible $n \times n$ matrix. Do the columns of A form a basis for $\mathbb{R}^n$?

The Standard Basis for $\mathbb{R}^n$

The columns $\mathbf{e}_1, \dots, \mathbf{e}_n$ of the $n \times n$ identity matrix form the standard basis for $\mathbb{R}^n$.

Example 3

Verify that $S = \{1, t, t^2, \dots, t^n\}$ is a basis for $\mathbb{P}_n$.

This basis is the standard basis for $\mathbb{P}_n$.

The Spanning Set Theorem

Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ be a set in a vector space V , and let $H = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\} = \text{Span } S$.

- If a vector $\mathbf{v}_k$ in S is a linear combination of the remaining vectors in S , then the set formed by removing $\mathbf{v}_k$ from S still spans H .
- If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H .

A Basis for $\text{Nul } A$

Our procedure for finding a set that spans $\text{Nul } A$ automatically produces a linearly independent set, and a linearly independent spanning set is a basis.

Thus, we already know how to find a basis for $\text{Nul } A$.

A Basis for Col A

The pivot columns of a matrix A form a basis for Col A .

A Basis for Row A

If two matrices A and B are row equivalent, their row spaces are the same.

If B is in echelon form, the nonzero rows of B form a basis for the row space of B and also the row space of A .

Example 4

Given that A and B are row equivalent, find bases for the row space of A , the column space of A , and the null space of A .

$$A = \begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ -2 & 3 & -3 & -3 & -4 \\ 4 & -6 & 9 & 5 & 9 \\ -2 & 3 & 3 & -4 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ 0 & 0 & 3 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Example 5

Find a basis for the set of all vectors of the form

$$\begin{bmatrix} a - 2b + 5c \\ 2a + 5b - 8c \\ -a - 4b + 7c \\ 3a + b + c \end{bmatrix}$$

Example 6

Let V and W be vector spaces, let $T: V \rightarrow W$ be a linear transformation, and let $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ be a subset of V .

Show that if $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is linearly dependent in V , then the set of images $\{T\{\mathbf{v}_1\}, \dots, T\{\mathbf{v}_p\}\}$ is linearly dependent in W .

Alternate Views of Bases

A basis is a minimal spanning set for a vector space.

A basis is a maximal linearly independent set in a vector space.
